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Corresponding author(s): Isabelle Régner, Pascal Huguet

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Reporting Summary

Nature Research wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Research policies, see Authors & Referees and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☐	☒ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	The Gender-Science IAT and questionnaire data were gathered on the WEB from Greenwald, Banaji, and Nosek's Project Implicit web-platform.
Data analysis	IBM SPSS statistics 24 with PROCESS macro (Hayes, 2013).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors/reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

Datafile and codebook are available on on OSF (https://osf.io/umf62/).

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☒ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	Data of the present study were all quantitative data obtained through both explicit (questionnaire) and implicit (Implicit Association Test) measures . Each committee's decision outcomes for years 1 and 2 (i.e., the number of men and women selected and in the candidate pools) were also used to compute an adverse impact (AI) ratio that compares the pass rates of men and women while taking into account the number of men and women in the applicant pool.
Research sample	The study involves human research participants (254 men, 154 women, 6 unspecified) aged from 35 to 64. The age information is not included in our database for ethical reasons related to anonymity : The governing body asked us not to keep this information that, combined with other information (e.g., gender, disciplines, section membership) could help identify the respondents. Participants were all members of the National Committee for Scientific Research, which plays a key role in French science, who were encouraged by the governing body to participate in a study on women's underrepresentation in science. This national committee is a collective body comprised of a general Scientific Board, 10 Institute Scientific Boards, and 40 specialized committees that cover the entire scientific spectrum (from Math and Physical Theories to Political Sciences). As indicated in Figure 1's caption (main text), the committees' structure is ordered by the governing body from STEM fields (numbers from 1 to 20) to life/social sciences (numbers from 21 to 40). Members of these specialized committees (about 20 researchers per committee) meet three times a year typically during 3-4 days.
Sampling strategy	For the period of the study, the population of interest was the CNRS Evaluation Committees which includes 40 committees that represent the whole scientific spectrum from Maths and Physical Theories to Political Sciences. Participation in the study was made available to this entire population. Our final sample includes data from 39 of the 40 available committees. That committee-level data is derived from the scores and responses from 414 committee members (254 men, 154 women, 6 unspecified), which represents a bit more than 50% of the whole population. This participation rate is relatively high considering that participants come from the real world and their decisions would have real consequences for the candidates' actual careers. In contrast to people participating in mock-hiring scenarios, real committee members may be reluctant to report on their attitudes and beliefs about women's underrepresentation in science. And yet, our 50% rate of participation is greater than the 30-35% response rate in prior high profile experimental investigations of gender bias in selection decisions using academic samples.
Data collection	Each participant had free access to a computer throughout the day in a room where they worked alone (15 min in total) during their breaks and other free times. The participation of the committee members was limited to performing the Gender-Science IAT and filling out the questionnaires.
Timing	Participants responses to the questionnaire and IAT were collected from April to June 2010, and the governing body gave us access to each committee's decision outcomes for years 1 and 2 (i.e., the number of men and women selected and in the candidate pools) at the end of 2011.
Data exclusions	Data exclusions were made exclusively on raw data gathered with the Implicit Association Test (IAT) at the individual level (not at the Committee level). As indicated in the Methods and following usual guidelines (Nosek, Greenwald, & Banaji, 2005), IAT scores were disqualified for any of the following criteria suggestive of careless participation: 1) going too fast (<300 ms) on more than 10% of the total test trials, 2) 25% of responses too fast in any one of the critical blocks, 3) 35% too fast in any one of the practice blocks, 4) making more than 30% erroneous responses across the critical blocks, 5) 40% errors in any one of the critical blocks, 6) 40% errors across all of the practice blocks, or 7) 50% errors in any one of the practice blocks. These standards resulted in a disqualification rate of 2.8% .
Non-participation	For the period of the study, the population of interest was the CNRS Evaluation Committees which includes 40 committees that represent the whole scientific spectrum from Maths and Physical Theories to Political Sciences. Participation in the study was made available to this entire population. Our final sample includes data from 39 of the 40 available committees. That committee-level data is derived from the scores and responses from 414 committee members (254 men, 154 women, 6 unspecified), which represents a bit more than 50% of the whole population. This participation rate is relatively high considering that participants come from the real world and their decisions would have real consequences for the candidates' actual careers. In contrast to people participating in mock-hiring scenarios, real committee members may be reluctant to report on their attitudes and beliefs about women's underrepresentation in science. And yet, our 50% rate of participation is greater than the 30-35% response rate in prior high profile experimental investigations of gender bias in selection decisions using academic samples.
Randomization	The present study was a survey, not an experiment. We tested a regressor variable model (i.e., the residual gain score model) that uses Year 1 Adverse Impact as a covariate.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology
☒	☐ Animals and other organisms
☐	☒ Human research participants
☒	☐ Clinical data

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Human research participants

Policy information about studies involving human research participants

Population characteristics	Participants were 414 members (254 men, 154 women, 6 unspecified) of the National Committee for Scientific Research, which plays a key role in French science. To guarantee anonymity, no other information was available.
Recruitment	All the members of the National Committee for Scientific Research, which plays a key role in French science, were encouraged by the governing body to participate in a study on women's underrepresentation in science.
Ethics oversight	The study and potentially related ethical issues were approved by the CNRS governing body at the time of the current study, the Mission for Women's Integration, and the National Committee of Scientific Research (CoNRS General Secretary).

Note that full information on the approval of the study protocol must also be provided in the manuscript.